

Optimizing Automatic Differentiation in Julia: Comparison with Pytorch and Flux

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Abstract—This paper presents a custom AD engine built from scratch in Julia, implementing backward accumulation via a computational graph. To achieve competitive performance, we applied several low-level optimizations: im2col transformations for convolutional layers, strict type stability enforcement, Float32 precision, and memory-efficient in-place operations combined with loop vectorization (@simd) and bounds checking elimination (@inbounds). We benchmarked our optimized library against PyTorch and Flux on a standard CNN architecture with full FashionMNIST dataset. Results show that our optimized implementation completes three training epochs in 19 seconds, compared to 30 seconds for Flux and 35 seconds for PyTorch. Additionally, our engine’s memory allocation is only 30% of that required by Flux. This study demonstrates that Julia can deliver exceptional performance for custom deep learning tools.

Index Terms—Julia, CNN, optimization, PyTorch, Neural Networks, SDG, Auto Differentiation

I. INTRODUCTION

Deep learning architectures, particularly Convolutional Neural Networks (CNNs), demand massive computational resources and highly efficient gradient computation. Automatic Differentiation (AD) serves as the mathematical foundation of this process, enabling the dynamic and accurate computation of derivatives necessary for backpropagation. While established frameworks like PyTorch and Flux provide highly optimized, production-ready AD engines, their heavily abstracted nature often obscures the underlying performance bottlenecks and memory management challenges.

The Julia programming language presents a compelling environment for deep learning by solving the “two-language problem” – promising high-level, mathematically intuitive syntax combined with C-like execution speed. However, because Julia is dynamically typed, realizing its full performance potential requires deliberate architectural decisions. Naive implementations of neural network training in Julia frequently suffer from type instability, excessive memory allocation, and unoptimized loops. These issues invoke heavy Garbage Collector overhead and drastically degrade execution speed.

To investigate and overcome these fundamental bottlenecks, we developed a custom AD library from scratch, implementing backward accumulation over a dynamic computational graph. Building a custom engine provides granular control over the entire execution pipeline. This paper explores the

critical optimizations required to make a custom Julia AD engine highly performant. By applying techniques such as im2col transformations, Float32 precision enforcement, strict type stability, and memory-efficient in-place operations, we demonstrate how to effectively bypass the overhead.

II. OPTIMIZATION APPLIED IN OUR LIBRARY

- A. *Types instability*
- B. *im2col*
- C. *float32*
- D. *Weights initialization*
- E. *In-place operations*

III. COMPARISON WITH OTHER LIBRARIES

- A. *Time complexity*

TABLE I
MY NETWORK EFFICIENCY RESULTS

Trial	Time (s)	Allocs	Mem (GB)	Train Acc (%)	Test Acc (%)
1	21,74	$4,24 \times 10^7$	2,47	87,74	86,53
2	21,99	$4,24 \times 10^7$	2,47	87,78	86,53
3	22,02	$4,24 \times 10^7$	2,47	87,87	86,67
4	21,50	$4,24 \times 10^7$	2,47	87,74	86,62
5	21,81	$4,24 \times 10^7$	2,47	87,64	86,57

TABLE II
FLUX EFFICIENCY RESULTS

Trial	Time (s)	Allocs	Mem (GB)	Train Acc (%)	Test Acc (%)
1	42,47	$6,01 \times 10^7$	26,09	88,10	86,88
2	29,12	$1,25 \times 10^7$	23,83	88,04	86,65
3	28,96	$1,25 \times 10^7$	23,83	87,44	86,50
4	26,76	$1,25 \times 10^7$	23,83	87,14	85,98
5	27,62	$1,25 \times 10^7$	23,83	87,94	86,52

- B. *Memory usage*

IV. CONCLUSION

Julia is fast.

REFERENCES

- [1] G. Eason, B. Noble, and I. N. Sneddon, “On certain integrals of Lipschitz-Hankel type involving products of Bessel functions,” *Phil. Trans. Roy. Soc. London*, vol. A247, pp. 529–551, April 1955.

TABLE III
PYTORCH EFFICIENCY RESULTS

Trial	Time (s)	Train Acc (%)	Test Acc (%)
1	32.94	86.93	85.75
2	32.52	87.74	86.79
3	32.47	87.47	86.01
4	32.21	87.56	86.00
5	33.09	87.64	86.66